A PARITY LEMMA FOR THE CODIFFERENT OF REAL CYCLOTOMIC FIELDS

ABSTRACT. Let $n = n_1 n_2 n_3$ with $n_1, n_2, n_3 > 1$ odd and pairwise coprime, and let $F_m = \mathbb{Q}(\zeta_m)^+$. We prove a congruence modulo 2 between the traces of an element of the codifferent of F_n to the three subfields $F_{n_i n_j}$. It shows, without any use of positivity, that no element of the codifferent has the products $a_i a_j$ of trace-one elements $a_i \in F_{n_i}$ as its three traces. Consequently, if each of the fields F_{pq}, F_{pr}, F_{qr} (for distinct odd primes p, q, r) has only products as its level-two elements and $\{3,5\} \not\subseteq \{p,q,r\}$, then the minimal polynomial of $2\cos(2\pi/pqr)$ is not 2-interlacing. We record what is known about this rigidity hypothesis, how the pair $\{3,5\}$ escapes the congruence, and an explicit family of non-products in F_{3q}.

1. Setting

For $n \geq 3$ let Ψ_n be the minimal polynomial of $\alpha_n = 2\cos(2\pi/n)$, of degree $d = \varphi(n)/2$, and let $F_n = \mathbb{Q}(\alpha_n) = \mathbb{Q}(\zeta_n)^+$, so that $O_{F_n} = \mathbb{Z}[\alpha_n]$ and the codifferent is $\mathcal{D}_{F_n}^{-1} = \mathbb{Z}[\alpha_n]^\vee = \Psi_n'(\alpha_n)^{-1}\mathbb{Z}[\alpha_n]$. A polynomial $g \in \mathbb{Z}[x]$ of degree $d-1$ with leading coefficient $j \geq 1$ interlaces Ψ_n (its roots are real and lie strictly between consecutive roots of Ψ_n) if and only if $\gamma = g(\alpha_n)/\Psi_n'(\alpha_n)$ is a totally positive element of $\mathcal{D}_{F_n}^{-1}$ with $\mathrm{Tr}_{F_n/\mathbb{Q}}\,\gamma = j$: this is [Y, Prop. 3.5] for monic g and the remark following its proof in general.

Definition 1.1. A *level-j element* of a totally real field F is a totally positive $\gamma \in \mathcal{D}_F^{-1}$ with $\mathrm{Tr}_{F/\mathbb{Q}}\,\gamma = j$. For $F = F_n$ we write $m(n)$ for the least j for which a level-j element exists; thus Ψ_n is j-interlacing if and only if F_n has a level-j element.

If $E \subseteq F$ then $\mathrm{Tr}_{F/E}$ maps level-j elements of F to level-j elements of E: total positivity is clear, and for $\gamma \in \mathcal{D}_F^{-1}$ and $e \in O_E$ we have $\mathrm{Tr}_{E/\mathbb{Q}}(e\,\mathrm{Tr}_{F/E}\,\gamma) = \mathrm{Tr}_{F/\mathbb{Q}}(e\gamma) \in \mathbb{Z}$. Hence m can only grow from a subfield to the field.

Theorem 1.2 ([Y, Thm. 1.2])**.** *Let n be squarefree, not a prime or twice a prime, with $n > 20$ and $n \neq 30$. Then Ψ_n is non-interlacing, i.e. $m(n) \geq 2$.*

The proof in [Y] rests on [BF, Cor. 2.2]: a definite lattice with an automorphism of characteristic polynomial Φ_m, m squarefree, $m \neq p, 2p$ and $\varphi(m) > 8$, does not represent 2. For odd squarefree n with at least two prime factors, Theorem 1.2 applies to every $n \neq 15$; and $m(15) = 1$.

The natural next question is whether $m(pqr) \geq 3$ for distinct odd primes p, q, r (with $\{3,5\} \not\subseteq \{p,q,r\}$, see Section 5.3). Computations in this repository show that F_n has no level-2 element for $n = 231$, 273, 357, 385, 399, 429, the six such n below 455 (`results/higher_interlacing_cyclotomic.md`). This note isolates a mod-2 obstruction which reduces part of the question to two primes.

2. Arrays

Let $n \geq 3$ be odd, $M_n = \mathbb{Q}(\zeta_n)$ with $\zeta_n = e^{2\pi i/n}$, and $F_n = M_n \cap \mathbb{R}$. For $\delta \in M_n$ define the *array*

$$c_\delta \colon \mathbb{Z}/n \to \mathbb{Q}, \qquad c_\delta(x) = \mathrm{Tr}_{M_n/\mathbb{Q}}(\delta\,\zeta_n^x).$$

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Lemma 2.1. (1) *The map $\delta \mapsto c_\delta$ is injective and $\mathbb{Q}$-linear.*
(2) *If $\delta \in F_n$ then $c_\delta(-x) = c_\delta(x)$ and $c_\delta(0) = 2\,\mathrm{Tr}_{F_n/\mathbb{Q}}\,\delta$.*
(3) *If $\delta \in \mathcal{D}_{F_n}^{-1}$ then c_δ takes values in $\mathbb{Z}$.*
(4) $\sum_{x\in\mathbb{Z}/n} c_\delta(x) = 0$.

Proof. (1) The trace form of $M_n/\mathbb{Q}$ is non-degenerate and the ζ_n^x span M_n. (2) For real δ, $c_\delta(-x)$ is the complex conjugate of $c_\delta(x)$, which is rational; and $\mathrm{Tr}_{M_n/\mathbb{Q}}\,\delta = 2\,\mathrm{Tr}_{F_n/\mathbb{Q}}\,\delta$. (3) The different $\mathcal{D}_{M_n} = \mathcal{D}_{F_n}\mathcal{D}_{M_n/F_n}$ is contained in $\mathcal{D}_{F_n}O_{M_n}$, so $\mathcal{D}_{F_n}^{-1} \subseteq \mathcal{D}_{M_n}^{-1}$, and $\mathrm{Tr}_{M_n/\mathbb{Q}}(\delta\zeta_n^x) \in \mathbb{Z}$ for $\delta \in \mathcal{D}_{M_n}^{-1}$ since $\zeta_n^x \in O_{M_n}$. (4) $\sum_x \zeta_n^x = 0$. □

Remark 2.2. Writing $\sigma_k(\zeta_n) = \zeta_n^k$, one has $\sum_x c_\delta(x)\zeta_n^{-lx} = n\,\sigma_l(\delta)$ for $l \in (\mathbb{Z}/n)^\times$ and 0 for non-units l, so $\delta \in F_n$ is totally positive if and only if this discrete Fourier transform is positive on the units. Also $c_\delta(x-y) = \mathrm{Tr}_{M_n/\mathbb{Q}}(\delta\,\zeta_n^x\overline{\zeta_n^y})$ is the Gram matrix of the ideal lattice $(\mathbb{Z}[\zeta_n], \mathrm{Tr}_{M_n/\mathbb{Q}}(\delta u\bar{v}))$ in the basis of powers of ζ_n; cf. [BF, Lemma 1.5]. Neither fact is used below.

Blocks. Now let $n = n_1\cdots n_k$ with $n_1,\dots,n_k > 1$ odd and pairwise coprime. Identify $\mathbb{Z}/n = \prod_i \mathbb{Z}/n_i$ by $x \mapsto (x \bmod n_i)_i$, and let $e_i \in \mathbb{Z}/n$ be the idempotent with $e_i \equiv 1 \pmod{n_i}$ and $e_i \equiv 0 \pmod{n_j}$ for $j \neq i$. Then $\zeta_{(i)} = \zeta_n^{e_i}$ is a primitive n_i-th root of unity and $\zeta_n^x = \prod_i \zeta_{(i)}^{x_i}$. For a non-empty $S \subseteq \{1,\dots,k\}$ put $n_S = \prod_{i\in S} n_i$ and $\zeta_S = \prod_{i\in S}\zeta_{(i)}$, a primitive n_S-th root of unity generating $M_{n_S} \subseteq M_n$. *Arrays of elements of M_{n_S} are taken with respect to ζ_S*, as functions on $\prod_{i\in S}\mathbb{Z}/n_i$. (Another choice of primitive root composes the array with multiplication by a unit; nothing below depends on it.)

Lemma 2.3. *Let $\delta \in M_n$.*
(1) (Line sums.) *For every i and every x, $\sum_{t\in\mathbb{Z}/n_i} c_\delta(x + te_i) = 0$; that is, c_δ sums to zero along every line parallel to a coordinate axis.*
(2) (Traces are restrictions.) *For non-empty S, $c_{\mathrm{Tr}_{M_n/M_{n_S}}\delta}(x_S) = c_\delta(x_S, 0)$, where $(x_S, 0)$ has coordinates x_i for $i \in S$ and 0 otherwise. If $\delta \in F_n$ then $\mathrm{Tr}_{M_n/M_{n_S}}\,\delta = \mathrm{Tr}_{F_n/F_{n_S}}\,\delta$.*
(3) (Products are tensors.) *Let $\{1,\dots,k\} = S \sqcup S'$. If $a \in M_{n_S}$ and $b \in M_{n_{S'}}$, then $c_{ab}(x_S, x_{S'}) = c_a(x_S)\,c_b(x_{S'})$.*

Proof. (1) The sum is $\mathrm{Tr}_{M_n/\mathbb{Q}}\big(\delta\zeta_n^x\sum_t \zeta_{(i)}^t\big) = 0$. (2) For $w \in M_{n_S}$, $\mathrm{Tr}_{M_{n_S}/\mathbb{Q}}(w\,\mathrm{Tr}_{M_n/M_{n_S}}\,\delta) = \mathrm{Tr}_{M_n/\mathbb{Q}}(w\delta)$; take $w = \zeta_S^{x_S} = \zeta_n^{(x_S,0)}$. Restriction $\mathrm{Gal}(M_n/M_{n_S}) \to \mathrm{Gal}(F_n/F_{n_S})$ is injective, since an element of the kernel fixes F_n, hence is the identity or complex conjugation, and complex conjugation moves ζ_S ($n_S \geq 3$); both groups have order $\varphi(n)/\varphi(n_S)$. (3) $M_n = M_{n_S}\otimes_{\mathbb{Q}} M_{n_{S'}}$ since $\varphi(n) = \varphi(n_S)\varphi(n_{S'})$, and the trace of a tensor product of elements is the product of the traces. □

Remark 2.4 (Composita)**.** Let $n = n'n''$ with n', n'' coprime and a, b level-j', level-j'' elements of $F_{n'}$, $F_{n''}$. The fields $F_{n'}$, $F_{n''}$ have coprime discriminants, so $O_{F_{n'}F_{n''}} = O_{F_{n'}} \otimes O_{F_{n''}}$ and $ab \in \mathcal{D}_{F_{n'}F_{n''}}^{-1} \subseteq \mathcal{D}_{F_n}^{-1}$. As $[F_n : F_{n'}F_{n''}] = 2$, the element ab has level $2j'j''$ in F_n; hence $m(n'n'') \leq 2\,m(n')\,m(n'')$. We call these elements *products*. By Lemma 2.3(2),(3), if $\mathrm{Tr}_{F_{n''}/\mathbb{Q}}\,b = 1$ then $\mathrm{Tr}_{F_n/F_{n'}}(ab) = 2a$, because both have the array $c_a(x')\,c_b(0) = 2c_a(x')$.

3. The parity lemma

Throughout this section $n = n_1n_2n_3$ with $n_i > 1$ odd and pairwise coprime, and $A = \mathbb{Z}/n_1 \times \mathbb{Z}/n_2 \times \mathbb{Z}/n_3$. Let $\mathcal{A}$ be the set of functions $c\colon A \to \mathbb{Z}$ with $c(-x) = c(x)$ and zero sums along all lines parallel to the axes. By Lemmas 2.1 and 2.3, $c_\delta \in \mathcal{A}$ for every $\delta \in \mathcal{D}_{F_n}^{-1}$.

A *half-system* of $\mathbb{Z}/m$ (m odd) is a subset $T \subseteq \mathbb{Z}/m \setminus \{0\}$ with $\mathbb{Z}/m \setminus \{0\} = T \sqcup (-T)$; if T is one, so is $-T$. Fix half-systems T_i of $\mathbb{Z}/n_i$.

Lemma 3.1. *For every $c \in \mathcal{A}$,*

$$\sum_{(y,z)\in T_2\times T_3} c(0,y,z) + \sum_{(x,z)\in T_1\times(-T_3)} c(x,0,z) + \sum_{(x,y)\in T_1\times T_2} c(x,y,0) \equiv 0 \pmod 2.$$

Proof. Call the three sums X, Y, Z. For signs $\varepsilon \in \{\pm\}^3$ let $\Lambda(\varepsilon) = \sum c(x,y,z)$ over $(x,y,z) \in \varepsilon_1 T_1 \times \varepsilon_2 T_2 \times \varepsilon_3 T_3$. Since c is even, $\Lambda(-\varepsilon) = \Lambda(\varepsilon)$. Since the line sums vanish and $\mathbb{Z}/n_1 = \{0\} \sqcup T_1 \sqcup (-T_1)$, summing over lines in the first direction gives $\Lambda(+,\varepsilon_2,\varepsilon_3) + \Lambda(-,\varepsilon_2,\varepsilon_3) = -\sum_{\varepsilon_2 T_2 \times \varepsilon_3 T_3} c(0,y,z)$, and similarly in the other two directions. Taking $(\varepsilon_2,\varepsilon_3) = (+,+)$ in the first direction, $(\varepsilon_1,\varepsilon_3) = (+,-)$ in the second and $(\varepsilon_1,\varepsilon_2) = (+,+)$ in the third, and using $\Lambda(-,+,+) = \Lambda(+,-,-)$,

$$\begin{aligned}\Lambda(+,+,+) + \Lambda(+,-,-) &= -X,\\ \Lambda(+,+,-) + \Lambda(+,-,-) &= -Y,\\ \Lambda(+,+,+) + \Lambda(+,+,-) &= -Z.\end{aligned}$$

Adding, $X + Y + Z = -2\big(\Lambda(+,+,+) + \Lambda(+,-,-) + \Lambda(+,+,-)\big)$. □

Definition 3.2. Let $m_1, m_2 > 1$ be odd and coprime, and let $C\colon \mathbb{Z}/m_1 \times \mathbb{Z}/m_2 \to \mathbb{Z}$ be even, with zero line sums and with even-valued axes $C(x,0)$, $C(0,y)$. Its *corner parity* is

$$\tau(C) = \sum_{(x,y)\in T\times T'} C(x,y) \bmod 2$$

for half-systems T, T' of $\mathbb{Z}/m_1$, $\mathbb{Z}/m_2$. For $\beta \in \mathcal{D}^{-1}_{F_{m_1 m_2}}$ whose traces to F_{m_1} and F_{m_2} have even-valued arrays we put $\tau(\beta) = \tau(c_\beta)$; by Lemma 2.3(2) the axes of c_β are the arrays of these two traces.

Lemma 3.3. *$\tau(C)$ does not depend on the half-systems.*

Proof. Two half-systems differ by replacing some elements t by $-t$. Replacing $t \in T'$ by $-t$ changes the sum by $\sum_{x\in T}\big(C(x,-t) - C(x,t)\big) \equiv \sum_{x\in T}\big(C(-x,t) + C(x,t)\big) = \sum_{x\neq 0} C(x,t) = -C(0,t) \equiv 0$, using $C(x,-t) = C(-x,t)$ and the zero line sum through $(0,t)$. The same holds for T. □

Theorem 3.4 (Parity lemma)**.** *Let $\delta \in \mathcal{D}^{-1}_{F_n}$ and suppose that the arrays of $\mathrm{Tr}_{F_n/F_{n_i}}\delta$ are even-valued for $i = 1,2,3$. Then*

$$\tau\big(\mathrm{Tr}_{F_n/F_{n_2 n_3}}\delta\big) + \tau\big(\mathrm{Tr}_{F_n/F_{n_1 n_3}}\delta\big) + \tau\big(\mathrm{Tr}_{F_n/F_{n_1 n_2}}\delta\big) \equiv 0 \pmod 2.$$

Proof. Let $c = c_\delta \in \mathcal{A}$. By Lemma 2.3(2) and transitivity of the trace, the restriction of c to the coordinate plane through 0 orthogonal to the i-th axis is the array of $\mathrm{Tr}_{F_n/F_{n_j n_k}}\delta$, and its axes are the arrays of the traces to F_{n_j} and F_{n_k}, which are even-valued by hypothesis. Each restriction is even with zero line sums, so its corner parity is defined. The three sums of Lemma 3.1 compute these corner parities with the half-systems (T_2,T_3), $(T_1,-T_3)$ and (T_1,T_2), and by Lemma 3.3 the choice does not matter. □

Positivity plays no role: Theorem 3.4 is a statement about $\mathcal{D}^{-1}_{F_n}$ modulo 2.

Lemma 3.5. *Let $m_1, m_2 > 1$ be odd and coprime and let $a \in \mathcal{D}^{-1}_{F_{m_1}}$, $b \in \mathcal{D}^{-1}_{F_{m_2}}$ have trace 1. Then the traces of $ab \in F_{m_1 m_2}$ to F_{m_1}, F_{m_2} are $2a$, $2b$, and $\tau(ab) = 1$.*

Proof. By Lemma 2.3(3), $c_{ab} = c_a \otimes c_b$, with axes $2c_a$, $2c_b$ (Remark 2.4). By Lemma 2.1, c_a is integral and even, $c_a(0) = 2$ and $\sum_x c_a(x) = 0$; so $\sum_{x\in T} c_a(x) = \frac{1}{2}\sum_{x\neq 0} c_a(x) = -1$. Likewise for b, and $\tau(ab) = \big(\sum_T c_a\big)\big(\sum_{T'} c_b\big) = 1$. □

Corollary 3.6 (Product obstruction)**.** *Let $a_i \in \mathcal{D}^{-1}_{F_{n_i}}$ have trace 1 ($i = 1, 2, 3$). There is no $\delta \in \mathcal{D}^{-1}_{F_n}$ with*

$$\mathrm{Tr}_{F_n/F_{n_2n_3}}\delta = a_2a_3, \qquad \mathrm{Tr}_{F_n/F_{n_1n_3}}\delta = a_1a_3, \qquad \mathrm{Tr}_{F_n/F_{n_1n_2}}\delta = a_1a_2.$$

Proof. For such a δ, $\mathrm{Tr}_{F_n/F_{n_1}}\delta = \mathrm{Tr}_{F_{n_1n_2}/F_{n_1}}(a_1a_2) = 2a_1$, whose array $2c_{a_1}$ is even; likewise for n_2, n_3. By Lemma 3.5 the three corner parities are 1, and $1+1+1$ is odd, contradicting Theorem 3.4. □

Remark 3.7 (The number of blocks)**.** The proof of Lemma 3.1 works for any number k of blocks: walking from ε to $-\varepsilon$ by changing one sign at a time gives $\Lambda(\varepsilon)\big(1-(-1)^k\big)$ = an alternating sum of k face sums. For odd k this is a congruence modulo 2; for even k it is an exact linear relation. For $k = 2$ the relation reads $\sum_{T_1} c(x, 0) = \sum_{T_2} c(0, y)$, and both sides equal $-c(0, 0)/2$, so it carries no information. This matches the fact that products of level-one elements are level-two elements of F_{pq} (Remark 2.4), while three-fold products are ruled out by Corollary 3.6.

4. Three primes

Definition 4.1. For coprime odd $m_1, m_2 > 1$, the field $F_{m_1m_2}$ is *rigid at level two* if every level-two element of $F_{m_1m_2}$ is a product ab of level-one elements $a \in F_{m_1}$, $b \in F_{m_2}$.

Theorem 4.2. *Let $n = n_1n_2n_3$ with $n_i > 1$ odd and pairwise coprime. Suppose that for each pair $\{i, j\}$ the field $F_{n_in_j}$ has no level-one element and is rigid at level two. Then F_n has no element of level ≤ 2, i.e. $m(n) \geq 3$.*

Proof. Let δ have level $j \leq 2$. Its traces to the fields $F_{n_in_j}$ have level j (Section 1), so $j = 2$ and they are products: $\mathrm{Tr}_{F_n/F_{n_1n_2}}\delta = a_1a_2$, $\mathrm{Tr}_{F_n/F_{n_1n_3}}\delta = a_1'a_3$ and $\mathrm{Tr}_{F_n/F_{n_2n_3}}\delta = a_2'a_3'$. Tracing further to F_{n_1} and using Remark 2.4, $2a_1 = \mathrm{Tr}_{F_n/F_{n_1}}\delta = 2a_1'$, so $a_1 = a_1'$; likewise $a_2 = a_2'$ and $a_3 = a_3'$. This contradicts Corollary 3.6. □

Corollary 4.3. *Let p, q, r be distinct odd primes with $\{3, 5\} \not\subseteq \{p, q, r\}$. If F_{pq}, F_{pr} and F_{qr} are rigid at level two, then $m(pqr) \geq 3$: the minimal polynomial of $2\cos(2\pi/pqr)$ is not 2-interlacing.*

Proof. Each product of two of the primes is squarefree, neither prime nor twice a prime, exceeds 20 and differs from 30, so it has no level-one element by Theorem 1.2. Apply Theorem 4.2. □

Theorem 4.2 also applies to composite blocks; for example, the blocks $(15, p, q)$ need rigidity of F_{15p}, F_{15q} and F_{pq}, with products built from the level-one elements of F_{15}.

5. Status and examples

5.1. **Rigidity.** Table 1 is taken from `results/higher_interlacing_cyclotomic.md`, where the level-two elements were enumerated by towers of subfields. Table 2 sorts the level-two elements of the smallest fields F_{pq} by the type of their traces; it is reproduced by `scripts/parity_table.py` (about 15 seconds) and stored in `results/parity_level2.json`.

So no small F_{pq} is rigid, and the rigid pairs $\{7, 13\}$, $\{7, 17\}$, $\{7, 19\}$, $\{11, 13\}$ of Table 1 contain no triangle: completing one needs a further field such as F_{221} or F_{247} (degrees 96 and 108), for the triples $\{7, 13, 17\}$ and $\{7, 13, 19\}$. *At present Corollary 4.3 proves no case that was not already settled by computation.*

field	level-two elements	products	rigid
F_{77}	285	15	no
F_{91}	18	18	yes
F_{119}	24	24	yes
F_{133}	27	27	yes
F_{143}	30	30	yes

TABLE 1. Level-two elements of F_{pq} found in the computations of $m(pqr)$ for $pqr < 455$.

				non-products		
field	degree	level two	products	$\tau = 0$	$\tau = 1$	other
F_{15}	4	50	2	12	12	24
F_{21}	6	141	3	18	18	102
F_{33}	10	935	5	20	30	880
F_{35}	12	1470	6	36	84	1344
F_{39}	12	2214	6	48	24	2136

TABLE 2. Level-two elements β of F_{pq}. The middle columns count those with $\mathrm{Tr}_{F_{pq}/F_p}\beta = 2a$ and $\mathrm{Tr}_{F_{pq}/F_q}\beta = 2b$ for level-one elements a, b (products, and non-products by corner parity); "other" counts the rest.

5.2. **Gluing.** Theorem 3.4 also constrains the case where only two of the three fields are rigid.

Proposition 5.1. *Let $n = n_1n_2n_3$ be as in Section 3, and let δ be a level-two element of F_n whose traces to $F_{n_1n_3}$ and $F_{n_2n_3}$ are products of level-one elements. Then $\beta = \mathrm{Tr}_{F_n/F_{n_1n_2}}\delta$ is a level-two element with traces $2a_1$, $2a_2$ to F_{n_1}, F_{n_2} for level-one elements a_1, a_2; it is not a product, and $\tau(\beta) = 0$.*

Proof. Write the two traces as a_1a_3 and a_2a_3'; tracing to F_{n_3} gives $a_3 = a_3'$, and tracing to F_{n_i} gives $\mathrm{Tr}_{F_{n_1n_2}/F_{n_i}}\beta = 2a_i$. If β were a product it would be a_1a_2, contradicting Corollary 3.6. Theorem 3.4 with Lemma 3.5 gives $1 + 1 + \tau(\beta) \equiv 0$. □

Hence, for $n = pqr$ with F_{pr}, F_{qr} rigid, only the elements in the column $\tau = 0$ of Table 2 can occur as the trace to F_{pq} of a level-two element. For the six $n \le 429$ of Section 1 the computations show that F_n has no level-two element at all. Parity alone does not explain this, since every field of Table 2 has non-products with $\tau = 0$; positivity must be used as well.

5.3. **The pair $\{3, 5\}$.** The field F_{15} has eight level-one elements e, and each $2e$ has an even-valued array, so $\tau(2e) = 0$. For an odd prime $r \notin \{3, 5\}$ and a level-one element a_r of F_r, the element $\delta = e\,a_r \in F_{15}F_r$ has level 2 in F_{15r} (Remark 2.4). By Lemma 2.3, its traces to F_{15}, F_{3r} and F_{5r} are $2e$, $(\mathrm{Tr}_{F_{15}/\mathbb{Q}}\,e)\,a_r = a_r$ and $(\mathrm{Tr}_{F_{15}/F_5}\,e)\,a_r$, with corner parities 0, 1, 1, as Proposition 5.1 requires. This is how $\{3, 5\}$ passes the parity test, and together with Theorem 1.2 it gives $m(15r) = 2$ for every such r.

5.4. **Non-products in F_{3q}.**

Proposition 5.2. *Let $q > 3$ be prime, $K = F_{3q}$, $E = F_q \subset K$, $a = (2 - \alpha_q)/q$ and $\theta = (\zeta_3 - \zeta_3^{-1})(\zeta_q - \zeta_q^{-1}) \in K$. Then $\gamma_\pm = a(1 \pm \theta^{-1})$ lie in $\mathcal{D}_K^{-1}$, have trace 2, do not lie in E, and are totally positive if and only if $12\sin^2(\pi/q) > 1$, i.e. $q \in \{5, 7\}$.*

Proof. Put $\alpha = \zeta_3\zeta_q + \zeta_3^{-1}\zeta_q^{-1}$, a conjugate of α_{3q}, so that $O_K = \mathbb{Z}[\alpha]$. Its conjugate over E is $\alpha' = \zeta_3^{-1}\zeta_q + \zeta_3\zeta_q^{-1}$, and $\alpha - \alpha' = \theta$. Since $O_K = \mathbb{Z}[\alpha] \subseteq O_E[\alpha] \subseteq O_K$, the relative different is $\mathcal{D}_{K/E} = (\alpha - \alpha')O_K = \theta O_K$. Next, a is a level-one element of E: the prime $\mathfrak{p} = (2 - \alpha_q)$ of E is the only one above q, with $\mathfrak{p}^{(q-1)/2} = (q)$, and $E/\mathbb{Q}$ is tamely and totally ramified at q and unramified elsewhere, so $\mathcal{D}_E = \mathfrak{p}^{(q-3)/2}$ and a generates $\mathfrak{p}^{1-(q-1)/2} = \mathcal{D}_E^{-1}$; it is totally positive, and $\operatorname{Tr}_{E/\mathbb{Q}} a = (q - 1 - \operatorname{Tr}_{E/\mathbb{Q}} \alpha_q)/q = 1$ as $\operatorname{Tr}_{E/\mathbb{Q}} \alpha_q = -1$. Hence $\mathcal{D}_K^{-1} = \theta^{-1}\mathcal{D}_E^{-1}O_K$ contains a and a/θ. The non-trivial automorphism of K/E sends θ to $-\theta$, so $\operatorname{Tr}_{K/E} \theta^{-1} = 0$ and $\operatorname{Tr}_{K/\mathbb{Q}} \gamma_\pm = \operatorname{Tr}_{E/\mathbb{Q}}(2a) = 2$; and $\gamma_\pm \notin E$. At a real place of E with $\alpha_q \mapsto 2\cos(2\pi k/q)$, a is positive and $\theta^2 = (\alpha_3^2 - 4)(\alpha_q^2 - 4) = 12\sin^2(2\pi k/q)$, and θ takes the two values $\pm\sqrt{\theta^2}$ at the two places above it; so γ_+ and γ_- are positive at both if and only if $\theta^2 > 1$ there. The least value of $|\sin(2\pi k/q)|$ over $k \in (\mathbb{Z}/q)^\times$ is $\sin(\pi/q)$, and $12\sin^2(\pi/q)$ is 4.15, 2.26, 0.95 for $q = 5, 7, 11$ and decreases in q. □

The elements $\gamma_\pm$ have traces $2a$ to E and 2 to $F_3 = \mathbb{Q}$; their corner parity is 0 for $q = 5$ and 1 for $q = 7$. They account for some of the non-products of Table 2 in F_{15} and F_{21} only; the non-products in F_{33}, F_{35}, F_{39} are of other kinds.

5.5. **Open problems.**

(1) Prove that F_{pq} is rigid at level two for all primes $p, q \geq p_0$. With Corollary 4.3 this gives $m(pqr) \geq 3$ for $p, q, r \geq p_0$.
(2) For each of the finitely many non-rigid pairs expected, exclude the gluing of Proposition 5.1 uniformly in r, or find an exception like $\{3, 5\}$.
(3) For four or more primes one needs the minimal elements at the previous level, which are not products: products have level 4 in F_{pqr}, while $m(231) = 3$.

5.6. **Verification.** `tests/test_parity.py` checks Lemma 3.1 on random elements of $\mathcal{D}_{F_{105}}^{-1}$ for three orderings of the blocks $(3, 5, 7)$, Lemma 3.5 in F_{35}, the rows F_{15} and F_{21} of Table 2, and the positivity statement of Proposition 5.2 for $q = 5, 7, 11, 13$. Lemma 3.1 was also checked on random elements of $\mathcal{D}_{F_{231}}^{-1}$, and Proposition 5.2 for $q \leq 29$. The arrays are computed exactly as $c_\delta(x) = \operatorname{Tr}_{F_n/\mathbb{Q}}(\delta\, C_x(\alpha_n))$ with $C_x(2\cos t) = 2\cos(xt)$ (`noninterlacing/parity.py`).

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